pg 111 (Q4b)

$\tau = 420\ N/mm^2$

$G = 84000\ \frac{N}{mm^2}$

$C = 6$

$P = 1000\ N$

$y = 25\ mm$

design d, D, C, n_T, n, L_f
L_s, P, α

① Using static app

7.100 $\rightarrow \tau = K_s \frac{8PC}{\pi d^2}$

$K_s = \frac{4C-1}{4C-4} + \frac{0.615}{C}$

$d = 6.75$

std wire dia

$d = 7\ mm$

$C = \frac{D}{d}$

$D = 42\ mm$

7.100 $\rightarrow y = \frac{8PC^3 n}{Gd}$

$n = 8.5$

$n_T = n + 2$
$= 10.5$

④ $L_f = \underset{7.101}{L_s} + y + \ldots + 2)$

$L_f = dn + 2d + y + n(1)$

$L_f = \quad mm$

7.101 ⑤ $L_f = pn + 2d$

$\boxed{p = \quad}$

⑥ helix angle

$\alpha = \tan^{-1}\left(\frac{p}{\pi D}\right) = \_\_\_$